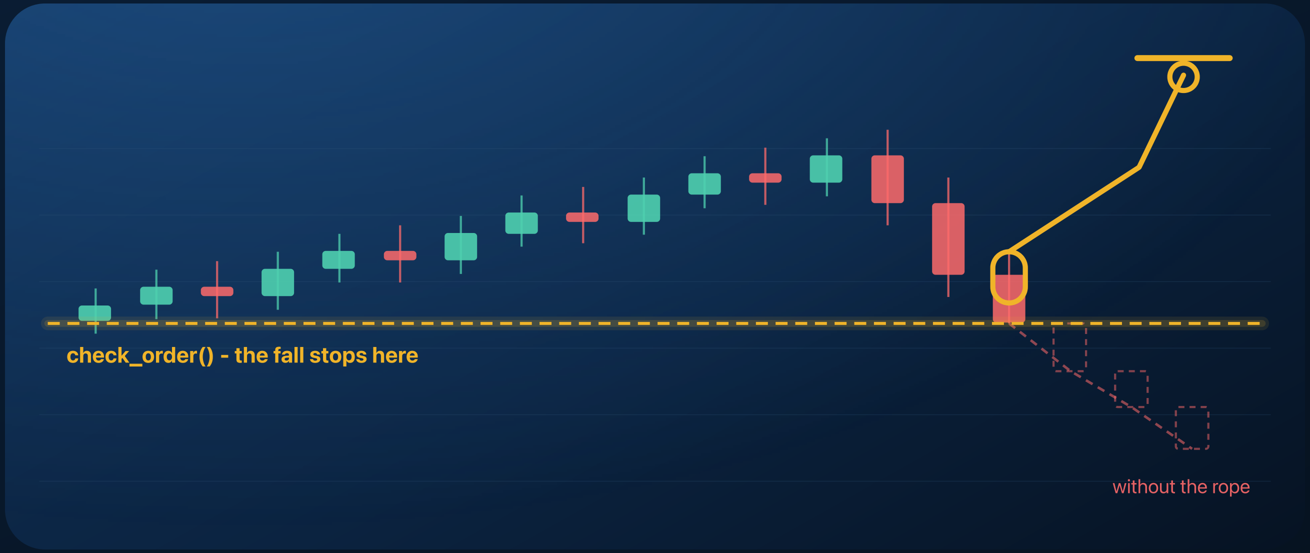
TEAM RAZORSEDGE

Autobelay

long premium, short leash



An auto belay catches a climber with nobody holding the other end. Here the brake is code.

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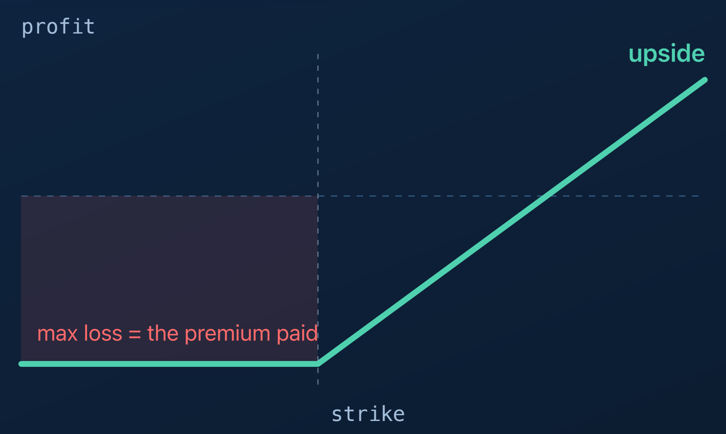
Alpaca AI Trading Agents Hackathon · Aug 28 – Sep 4, 2026

The thesis, in one sentence

Buy defined-risk, short-dated options premium when an open model can **name a reason**, and let deterministic code size it, stop it, and close it.



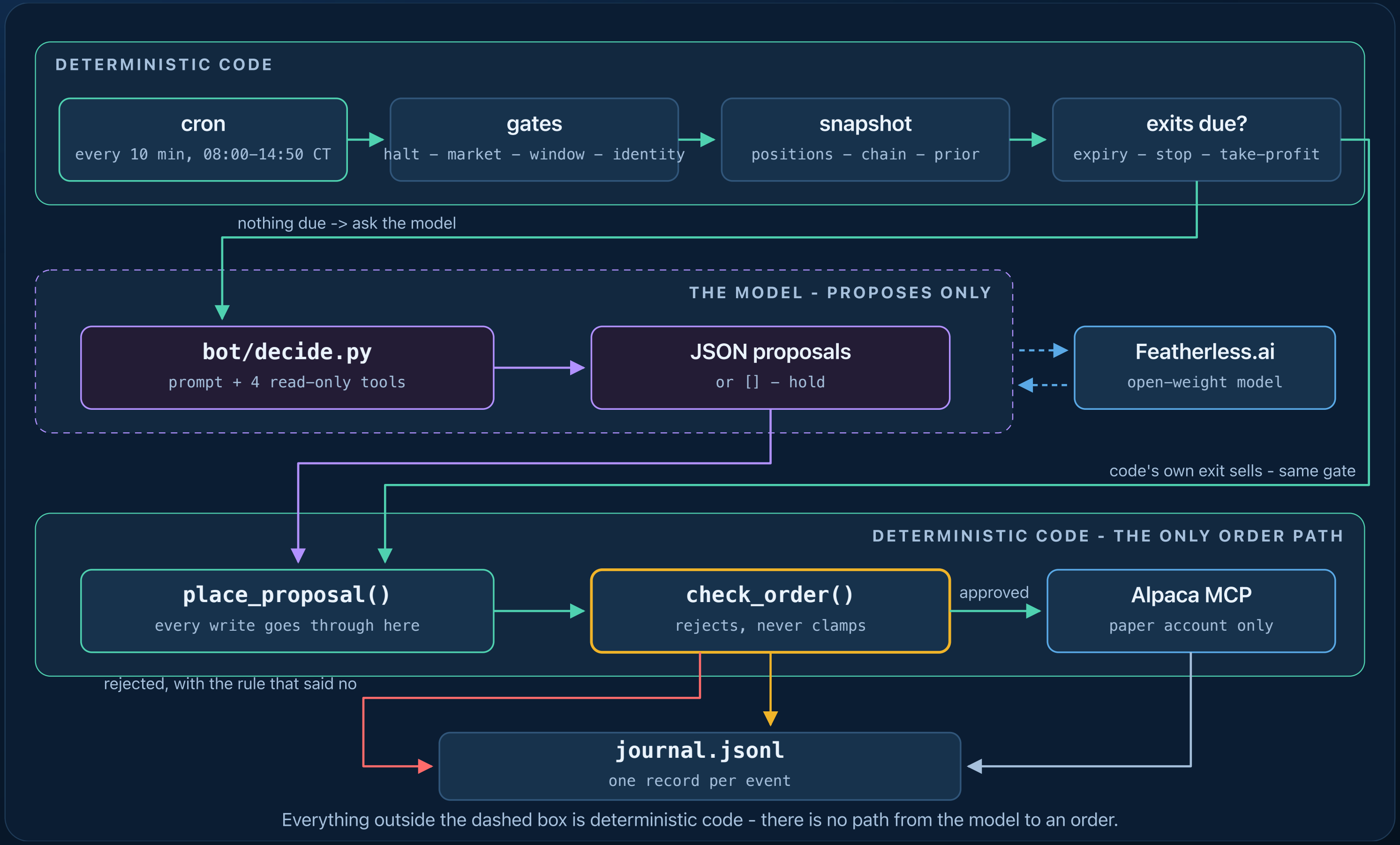
WHY LONG PREMIUM

Long premium only, so the worst case is known before entry, which is what lets every guardrail be a flat number.

Max loss is the debit paid. Nothing can be assigned against us.

The division of labour: the model reads the market and proposes; deterministic code sizes, stops and closes, and is the only thing that can reach an order. The next slide names every piece.

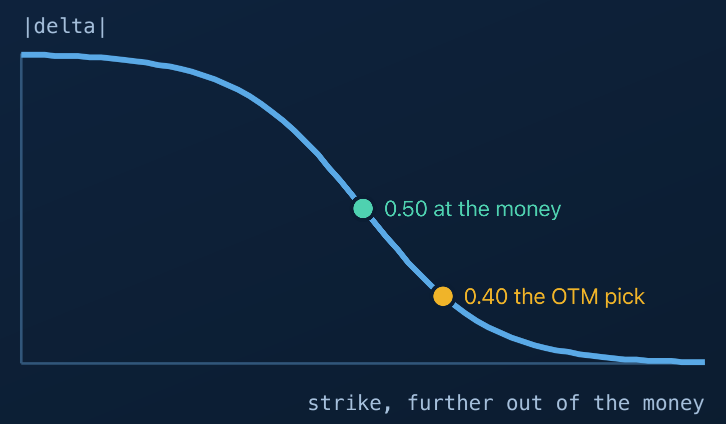
One cycle, every 10 minutes



The model gets one box. Its proposals and code's own exit sells pass the same gate.

Greeks: Alpaca's, ours as the backstop

We spent three days believing Alpaca's free feed carried no Greeks. We were wrong.



FROM ALPACA

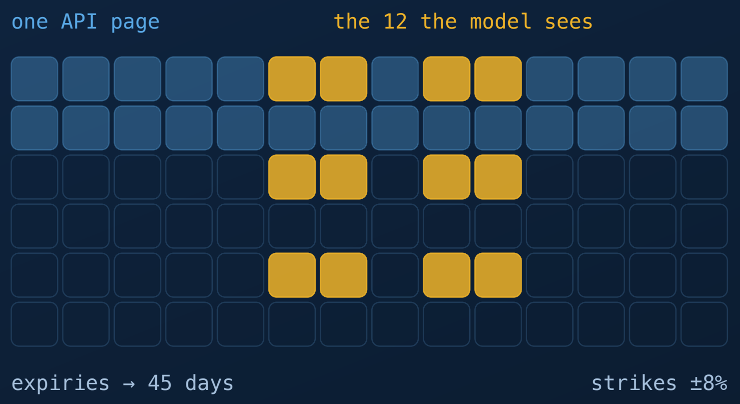
5,721 of 6,114 contracts in today's chain: one vendor surface, so a strike's put and call deltas agree.

BLACK-SCHOLES BACKSTOP

Our bisection solve covers the ~6% too thin to price. Every contract records which of the two priced it.

What the model may choose from

One API page is 500 contracts. On SPY, that is three days.



BEFORE

500 contracts, one page, out to 3 DTE. NVDA's twelve were all one strike.

NOW

2,912 contracts, three pages, out to 45. At-the-money *and* ~0.40 delta, three expiries, both sides, journaled every cycle.

A second opinion

Two crowds answer the same question: Kalshi's index-close market and the option chain's own implied odds. Both are priors to weigh rather than signals to copy, and no code acts on either.

KNOWING WHEN TO IGNORE IT

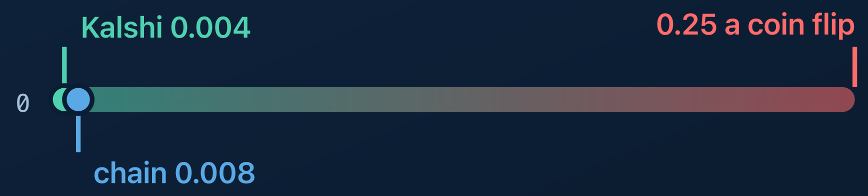
A barely-traded market quotes every bucket just as confidently, and nothing about its numbers looks broken. That is the point.

Volume and entropy gates withheld QQQ, **with the reason**, so "no second opinion today" is answerable rather than absent.

...and then we grade it

A prior nobody scores is decoration. Every night, the probabilities the model was handed are Brier-scored against what the market actually did - mean squared error of a probability forecast, where **0 is perfect and 0.25 is a coin flip**.

Brier score - lower is better



TUE SEP 1, JUDGED ACCOUNT

Kalshi **0.004**, the chain **0.008**. Priors the gate withheld score 0.006, shadow-graded, so we know it discarded nothing good.

One day, and a directional one. What is graded here is the *input* the model was handed, not the trade it made.

It investigates before it acts

```
research: get_bars({'symbol': 'SPY', 'timeframe': '15Min'})   -> 3334 chars
research: get_news({'symbols': 'NVDA', 'limit': 5})           -> 1770 chars
model output: [{"symbol":"NVDA260902P00220000","side":"buy","qty":10,
  "reason":"NVDA in a clear intraday downtrend (225 -> 218.6); the 220 put
            with 4 DTE, delta -0.58, aligns with the bearish momentum"}]
```

It may call four read-only tools, six times at most, and every call is journaled.
Nothing that places an order is reachable.

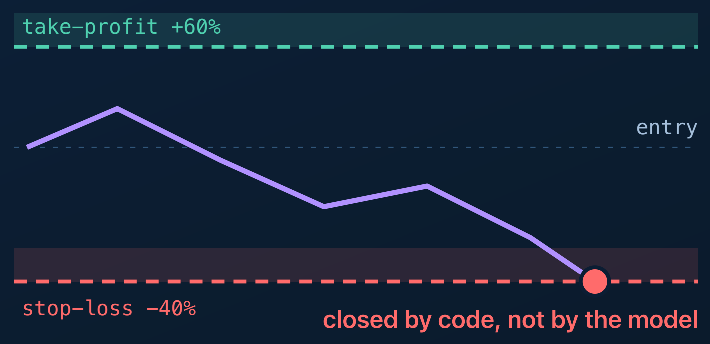
The leash

PER PROPOSAL

whitelist · ≤ $5,000 per position · ≤ 4 positions · ≤ 10 contracts · 2–45 DTE on entries · entries 09:45–15:15 ET · a resting buy counts as held

PER DAY

2% loss → flatten and halt. A `HALT` file is a global kill switch.



Everything is journaled

journal.jsonl

`one per account, append-only`

every decision, order, rejection, exit, tool call and config hash

THE BROWSER

Live journal viewer

`bot.wpmccormick.pw`

- journal_viewer.py, :8300 on the LAN
- -> cloudflared tunnel
- -> Cloudflare Access (email OTP)
- server-sent events, live follow
- read-only: no controls at all

THE DASHBOARD

Home Assistant

`MQTT auto-discovery`

- journal.log() -> fire-and-forget
- equity, P&L, halt, last decision
- kill switch + knobs, per-account
- sections layout: phone-friendly
- phone: problems only, never fills

THE INBOX

Hourly email

`mail_report.py`

- trades with the model's full reason
- CSVs attached, host local time
- silent when nothing traded
- one final send after the close

THE LOOP

eod_review.py

`the daily change`

- round trips from Alpaca's fills
- rejections grouped by rule
- Brier score on the priors
- critique by a model that did not trade

One file, four windows. Three are read-only views for humans; the fourth decides what changes tomorrow.

Watching it think, from anywhere

The journal streams to a browser at **bot.wpmccormick.pw**, published through a Cloudflare tunnel. There is no VPN and nothing to install.

```
14:30:17 official ✓ ORDER  sell 10 SPY260908P00760000 @ 4.0095
    Existing 8-DTE put at 760 strike is underwater (entry 4.57, current 3.92) with SPY at 761.28; thesis failed as market held above prior close despite prediction-market bearishness, and
    forced close approaches with 7 DTE remaining—cut loss before theta decay accelerates and backstop forces exit.
14:30:17 official ◀ end, 1 action(s)
14:40:09 mixed    ◆ PRIOR  SPY ref 7686.14  median 7637.5 (−0.63%)  P(above) 0.069  vol 11727 → shown
◆ PRIOR  QQQ ref 29456.97  median 29050 (−1.38%)  P(above) 0.058  vol 614.1 → shown
14:40:09 official ◆ PRIOR  SPY ref 7686.14  median 7637.5 (−0.63%)  P(above) 0.069  vol 11727 → shown
◆ PRIOR  QQQ ref 29456.97  median 29050 (−1.38%)  P(above) 0.058  vol 614.1 → shown
14:40:10 mixed    ▶ CYCLE  equity $100,151.92  day P&L 968.99  positions 0
14:40:10 official ▶ CYCLE  equity $100,209.83  day P&L 661.60  positions 0
```

The viewer found four bugs in two days, and none was a failing test: in each case the code did exactly what it had been told.

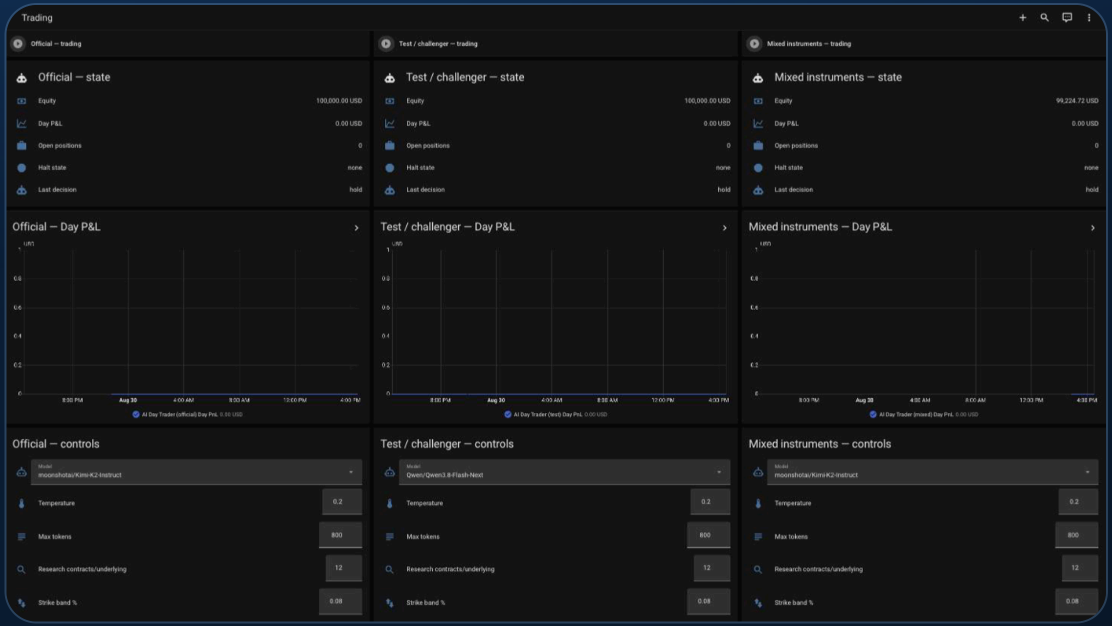
#174 #170

The feed cut the model's reasoning mid-word. Closing a +$1,340 winner, it named a *neighbouring strike*, three cycles running.

#171 #173

An unfilled limit sat invisible and the same thesis was bought again next door. A ten-minute-old position was proposed for exit, and every number it cited had moved *in its favour*.

One feed, three audiences



OPERATOR

Kill switch and knobs.

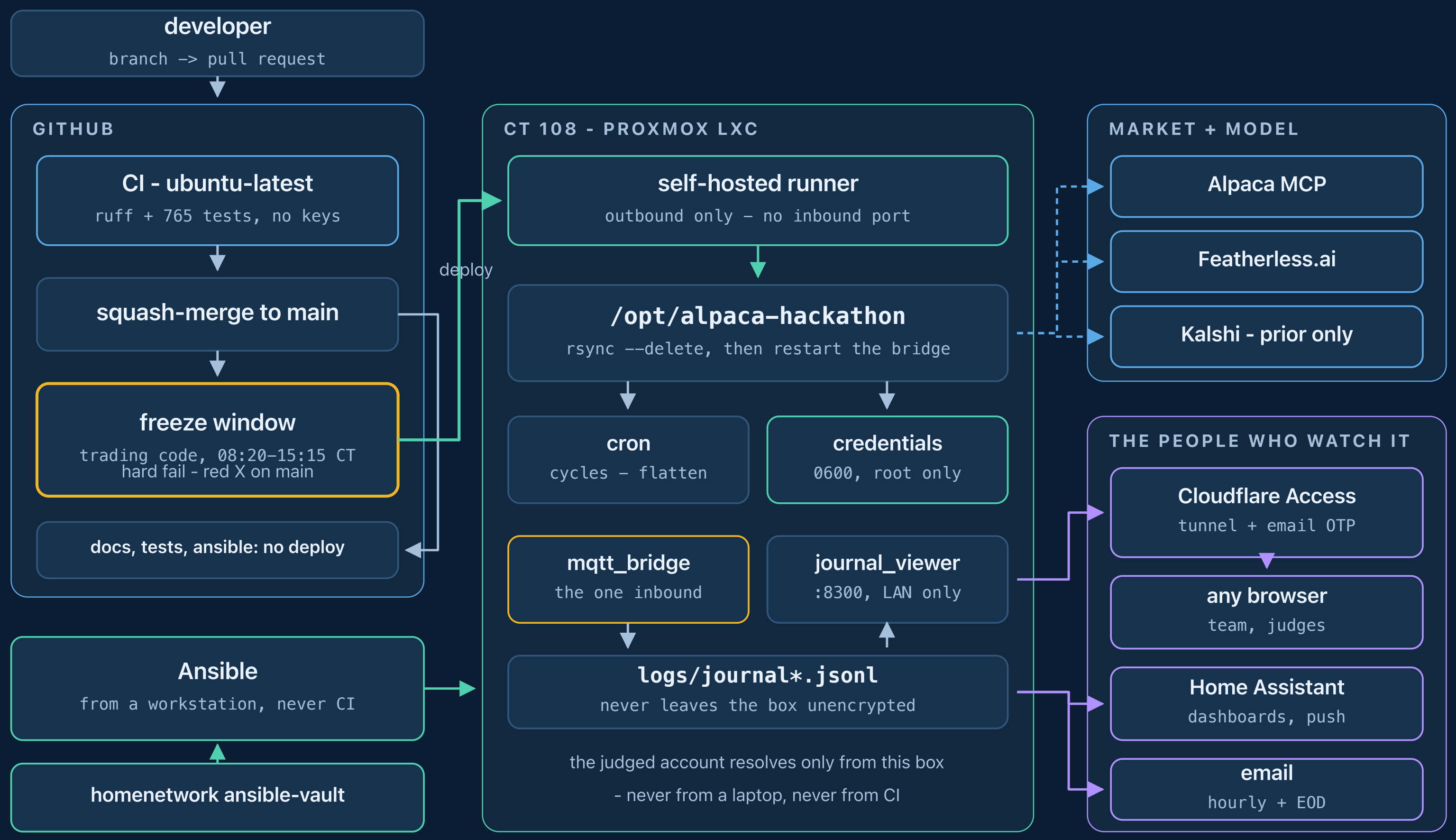
TEAM

The journal viewer, read-only by construction. Home Assistant has no per-entity permissions, so the separation is a different surface, not a second dashboard.

PHONE

Problems only. A fill sends nothing; *silence* does, when no cycle has run for 25 minutes.

Our own hardware, not a laptop



A merge deploys in about a minute. Nothing reaches the container inbound except the tunnel and the broker.

Built to be changed while it runs

765 tests, 2s	every one runs with **no API keys and no network**
Only Docker needed	tests, a full dry-run cycle, the docs site
Rehearse any time	`--dry-run` prints orders instead of sending; `--force` skips the market gates
No deploy needed	`override.py`: allowlisted knobs, hard caps stay git-only, and all of it expires at the close

Dozens of those tests exist because something broke. Each incident ends as a test.

The daily loop

16:05 `eod_review` → round trips · rejections by rule · Brier → a second model's read → one change → override or config PR → deployed before the open

The critique comes from a model that did *not* trade the day. That loop ran 160 pull requests and 125 deploys in six days.

Choosing the model, honestly

Featherless lists **21,912** models, **15,575** with tool calling on our plan. "Open weights" narrows nothing; our gates do, and none of them is a benchmark score.

THE GATES

Tool calling for the research loop · context for a 5k prompt that reaches **95k** tokens mid-research · a thinking-mode toggle, without which these models answer *empty* · a warm endpoint for a 10-minute cron · a price, so the day's spend is reported.

`scripts/verify_models.py` re-checks every one against the live catalog, so the claim cannot rot.

WHAT WE WILL NOT CLAIM

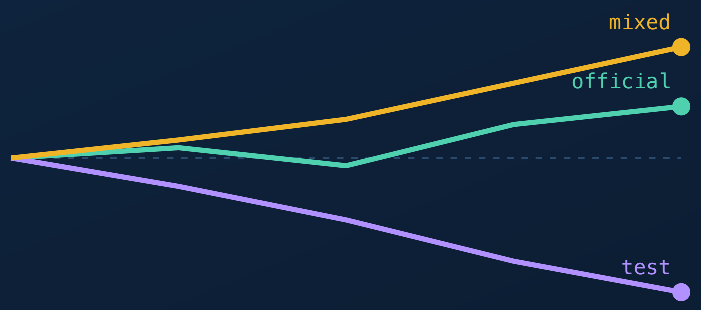
No bake-off picked the incumbent. Kimi was the first model confirmed to support tool calling, and it stayed. Our A/B varies the model *and* the tools *and* the learning loop, so it compares config bundles.

Saying so beats inventing a rationale for it.

The gate that discriminates is **instruction adherence**. `Llama-3.1-Hawkish-8B` — a finance fine-tune, cheaper than everything we run, clears every mechanical gate — proposed a 1-DTE contract in **3 of 3** runs and five actions against a four-position cap. Well-formed JSON, wrong content. It also found a real bug: the funnel permitted what the prompt forbade.

Real A/B, not a backtest

Four days is no backtest, and the free feed carries no historical options data, so the experiment runs live instead.



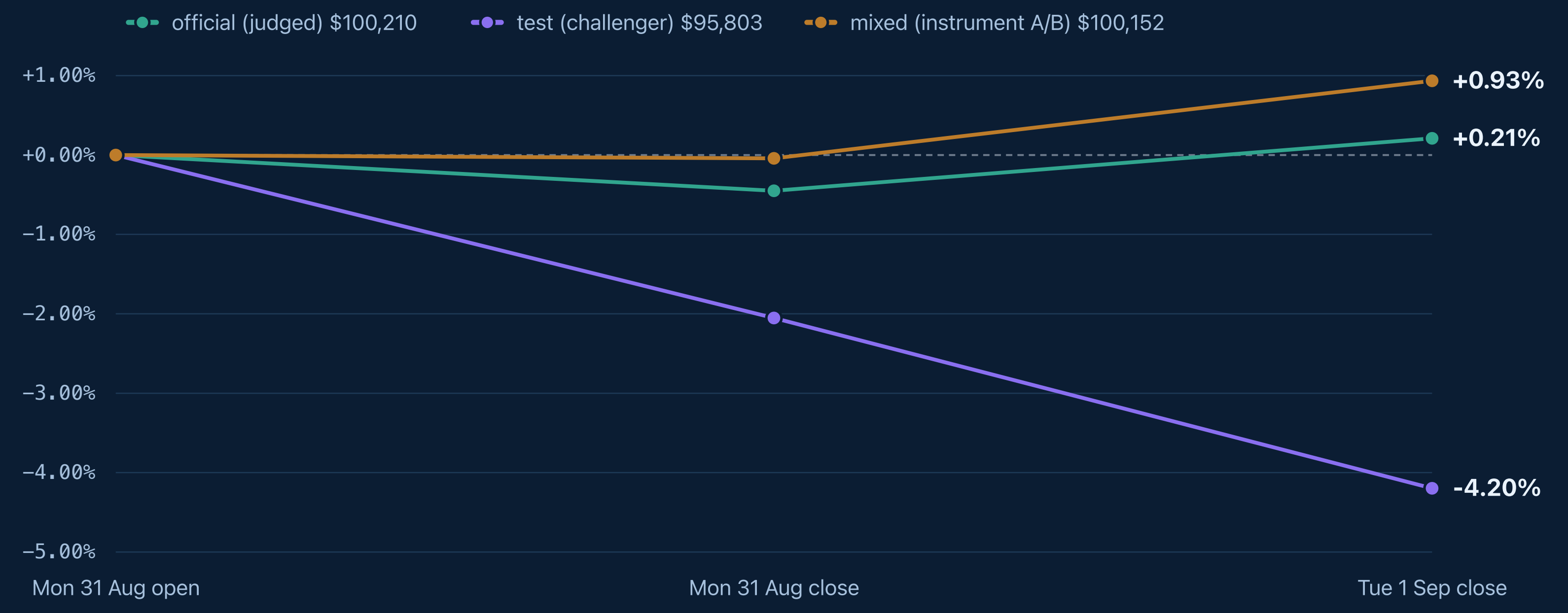
same market, same cadence, one difference each

THREE LIVE ACCOUNTS

Same box, same ten-minute cadence, same market, one deliberate difference each. `test` swaps the model and enables research and learning; `mixed` differs by exactly one key.

The winner is promoted into the official config by pull request.

Results (fill Thu Sep 3)



percent change from each account's open on 31 August - dollar equity at the close in the key

Judged account Mon open → Thu close: $___ → $___ (__%) · __ round trips · what didn't work: __

What's next

PROMOTE WHAT THE CHALLENGERS PROVED

Research tools, the prediction-market prior, the learning loop.

MULTI-LEG SPREADS

Once the gates cover assignment risk.

github.com/bill-mccormick-dg/alpaca-hackathon · MIT
Thanks to Alpaca, lablab.ai and Featherless AI.